# An M.Tech. Dissertation Report

**titled**

# CROSS-DATASET KNOWLEDGE DISTILLATION FOR CROP DISEASE DETECTION

**Submitted in partial fulfilment towards the award of the degree of**

## MASTER OF TECHNOLOGY

## in

## COMPUTER SCIENCE AND ENGINEERING WITH SPECIALIZATION IN DATA SCIENCE

## by


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# DECLARATION

I hereby declare that the work being presented in this dissertation preliminary report entitled "Cross-Dataset Knowledge Distillation for Crop Disease Detection" by me i.e. Ms. Yamini Cherukuri, bearing Admn. No: P24DS022 and submitted to the Computer Science and Engineering Department at Sardar Vallabhbhai National Institute of Technology, Surat; is an authentic record of my own work carried out during the period of July 2025 to December 2025 under the supervision of Dr. Sourajit Behera. The matter presented in this report has not been submitted by me in any other University/Institute for any cause.

Neither the source code there in, nor the content of the project report have been copied or downloaded from any other source. I understand that my result grades would be revoked if later it is found to be so.

_____

(Yamini Cherukuri)

# C E R T I F I C A T E

**This is to certify that the dissertation preliminary report entitled " Cross-Dataset Knowledge Distillation for Crop Disease Detection ", prepared and presented by Ms. Yamini Cherukuri, bearing Admn. No: P24DS022 of MTech. – II, Semester - III in Computer Science and Engineering with specialization in Data Science, at Department of Computer Science and Engineering of the Sardar Vallabhbhai National Institute of Technology, Surat is satisfactory.**

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# *Abstract*

*Agriculture continues to be the backbone of food security; however, plant disease detection in real- World conditions remains one of the biggest hurdles. Late diagnosis can often lead to reduced crop yield, poor quality, and significant economic losses to farmers. Though deep learning has greatly improved automated plant disease recognition, most previous methods a prioritize accuracy alone and overlook practical needs such as model efficiency, deployability, and robustness on field images. In this work, I present a lightweight yet high-performance plant disease classification framework that combines empirical transfer learning and knowledge distillation. Instead of blindly freezing or unfreezing layers, I systematically analyze and fine-tune the deeper layers of a pre-trained ResNet152 model to identify the optimal fine-tuning depth for disease recognition in potato, Apple and Strawberry leaves. To reduce the model size further and To improve inference efficiency, I distill knowledge from the fine-tuned ResNet152 teacher into compact ResNet18 student model. My approach effectively balances accuracy and computational cost, making it suitable for deployment on resource-constrained environments. By using On the PlantDoc potato dataset with a 75–25 train–test split, the proposed method shown substantial improvements over the baseline deep learning models with high efficiency. The distilled student network maintains strong performance with significantly fewer parameters. While highlighting the practicality of combining empirical finetuning with response-based knowledge distillation for real-world agricultural applications.*

***Keywords:*** Plant disease detection ,Deep Learning, Transfer Learning, Computer vision, Convolutional Neural Networs, Response-Based Knowledge Distillation

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# Chapter 1

# Introduction

## 1.1 Motivation

Agriculture remains the backbone in terms of food security and anchor for economic stability. Yet, diseases of plants remain major factors in lowering crop yield and degrading food quality due to significant losses incurred by farmers. Apples,Strawberry and Potatoes are some of the staple crops grown throughout the world, but are mostly vulnerable to foliar diseases.

Traditional manual disease inspection is slow, labor-intensive, and prone to human error under real-world field conditions where lighting, background clutter, and symptom variations make diagnosis difficult. Modern advances in deep learning and computer vision have presented promising prospects for automated plant disease detection; however, these models have yet to be widely deployed in practical agricultural settings due to issues such as limited training data, class imbalance, and the high computational cost of large pre-trained models.

Some of these challenges are addressed in the present work by investigating an effective approach that now combines **empirical fine-tuning** with **knowledge distillation**, where knowledge acquired from a powerful teacher model is transferred to a lightweight student model fit for scalable agricultural deployment.

## 1.2 Problem Statement

- Although deep learning models such as **ResNet152** achieve high accuracy in plant disease classification, they are computationally heavy and unsuitable for deployment on mobile and edge devices commonly used in agricultural environments.

- The New Plant Diseases dataset is clean and laboratory-generated, while real-field images from datasets such as *PlantDoc* contain noise, varied lighting, and complex backgrounds, creating a significant cross-domain generalization challenge.

- The teacher model (ResNet152) learns strong representations for apple, potato, and strawberry crops, but its large size and high computational demand make real-time field deployment impractical for farmers.
- There is a need to transfer the teacher's knowledge to a smaller, faster, and deployable model that can effectively classify potato leaf diseases in real-field conditions using minimal computational resources.
- This work addresses the problem by designing an **offline response-based Knowledge Distillation** framework to compress the large ResNet152 teacher into a lightweight **ResNet18** student model while preserving high classification performance.

If you want, I can also format this as a subsection, or add it under an Objective or Motivation chapter.

## 1.3 Our Contributions

The main contributions of this work are summarized as follows:

**Empirical Fine-Tuning Strategy**: I analyze and identify optimal layers of the ResNet152 model for fine-tuning in Apple,Strawberry,potato leaf disease classification, balancing accuracy and computational cost.

**Lightweight Knowledge Distillation Framework**: Response-based distillation means that the ResNet18 student model trains on the outputs of the teacher model, softening the predictions and shrinking parameters while retaining classification performance.

**Real-World Dataset Evaluation**: In this work, the proposed method is validated on the PlantDoc potato subset containing real-field images exhibiting considerable variations in lighting, background, and disease characteristics using a 75:25 train-test split.

**Deployment-Oriented Design**: The final student model offers improved efficiency, reduced size, and faster inference, suitable for mobile or edge-based agricultural applications.

## 1.4 Report Outline

This report is organized as follows:

- **Chapter II:** provides an overview of the relevant background and surveys existing research in plant disease detection and deep learning-based classification. It outlines the theoretical foundations and related works that form the basis for the methods adopted in this study.

- **Chapter III:** details the overall design of the proposed system and presents the analytical framework used to evaluate its components. It explains the architectural choices, workflow, and rationale behind each design decision.
- **Chapter IV:** describes the implementation details and the methodology followed to develop the proposed system. It explains the data processing steps, model training procedures, and the techniques applied to realize the design presented in the earlier chapters.
- **Chapter V:** presents the experimental results obtained from the proposed model and provides a detailed analysis of its performance across various evaluation metrics. It compares the outcomes with existing approaches and interprets the findings to demonstrate the effectiveness of the methodology.
- **Chapter VI:** summarizes the key findings of the study and highlights the contributions made through the proposed methodology. It also outlines potential directions for future research to further improve model performance and expand its real-world applicability.

# Chapter 2

# Literature Survey

The paper titled Efficient Plant Disease Classification Using ResNet-101 + Swarm-Optimized **[1]** Models presents the results obtained from using both machine learning and deep learning on the Clear Leaf Image Database for the classification of diseases present in apple, potato, and strawberry leaves. In the pipeline, the authors performed image preprocessing and segmentation, followed by feature extraction using ResNet-101, and classification through six different models, including Random Forest, ANN, LSTM, a baseline DeepCNN, and two swarm-optimized versions, namely, DO-DeepCNN and MFO-DeepCNN. Accuracy, sensitivity, and specificity were used to measure performance, with the best accuracy attained by MFODeep-CNN as 95.05%, closely tagged by DO-DeepCNN at 95.00%, while classical ML like RF and ANN presented minor overfitting with an accuracy of about 93%. A real soybean test further validated the robustness of the MFO-DeepCNN with an accuracy of approximately 94%. Important contributions of this work include the integration of evolutionary optimization with deep learning and the comprehensive comparison done between classical ML and DL approaches. Limitations inherent in this approach include the heavy computation of ResNet-101, no model compression or knowledge distillation, minor overfitting issues, and no consideration taken in regard to edge deployment constraints. These findings point towards the motivation for lightweight, efficient models, motivating the use of Knowledge Distillation (KD) to compress huge architectures such as ResNet-101/152 into compact models like ResNet-18 while maintaining high accuracy; it also falls in tune with what is intended by the current project.

The paper titled Leaf Disease Detection using Deep CNNs for Real-World Crop Diagnosis**[2]** focused on the task of detecting diseases in apple leaves under difficult real-field conditions. Using the New Plant Diseases Kaggle dataset with 70,000 training images, 17,000 validation samples, and 38 classes, the authors test several deep CNN architectures, including MobileNet, VGG, GoogLeNet, and a custom ResNet9 model. Their approach combined transfer learning, heavy data augmentation, and domain adaptation, training models on clean laboratory images while testing them on field images to measure the robustness to domain shift. Model ensembling was also used in order to improve the stability of the predictions. Among these compared models, ResNet9 outperformed others with 99.1% validation accu-

racy and 86.4% accuracy during field tests. This study provides valuable information about the influence of domain shift and clearly shows that deeper CNNs outperform shallower ones when applied to real-world agricultural imagery. Nevertheless, this work is limited by its concentration on one crop (apple) and also by the absence of model compression or knowledge distillation techniques. These two limitations create motivation to further explore whether it is possible for a lightweight student model, such as ResNet18, to approximate the performance of a much deeper teacher model such as ResNet152 on a similar task under significant domain shift—a goal in direct alignment with that of the present project.

The work titled "Improving Plant Disease Classification with Deep-Learning-Based Prediction Model Using Explainable Artificial Intelligence**[3]**" thus aims at the development of a highly accurate yet interpretable deep learning system for plant disease classification. Using the Kaggle New Plant Diseases dataset consisting of 87,000 images across 38 classes, the authors make use of transfer learning with EfficientNetB0 and fine-tune its last layers in order to attain better performance. The model is further supported by LIME-based explainable AI techniques that enable users to understand which regions of the leaf contribute to model decisions. This work also compares EfficientNetB0 against other CNN architectures and deploys the final model using the PlantCare mobile application. Its results are impressive, with 99.69% accuracy, 98.27% precision, and 98.26% recall. Key contributions include integrating explainability into plant disease detection and demonstrating practical mobile deployment. However, it is computationally heavy, without knowledge distillation or any form of compression, and optimized for accuracy rather than efficiency. Limitations here involve the need to have lighter, deployable models, which supports the motivation of the present project in using Knowledge Distillation-achieving EfficientNet-level performance with compact architectures like ResNet18.

The paper AI-IoT Smart Agriculture Pivot for Plant Disease Detection and Treatment**[4]** presents the design of an innovative, AIoT-based pivot irrigation system for automated detection of plant diseases and their real-time treatment in the agricultural field. The system design integrates IoT hardware, including cameras and sprayer units mounted on a pivot irrigation machine for continuous monitoring of crop health. The performance of a fine-tuned ResNet50 for disease classification across 11 classes is evaluated using a dataset of 25,940 images from tomato, potato, and bell pepper leaves. The performance of the model is very good, with test accuracy and precision, recall, and F1-scores of 99.8% and over 99.9%, respectively. The major contributions of this work are the successful integration of deep learning with smart irrigation technology and the mobile application developed for farmers to monitor and intervene remotely. On the downside, while ResNet50 performed exceptionally well, it is also extremely computationally heavy and may not be suitable for edge devices with low processing capabilities. Another flaw in the work is that, while transfer learning was adopted, model compression

techniques like Knowledge Distillation have not been applied. This provides a great opportunity for the work proposed here to demonstrate how a lightweight ResNet18 student model, trained through KD, can possibly achieve results that match those by ResNet50 while being much more efficient, scalable, and better suited for real agricultural deployment.

The work Knowledge Distillation: A Survey**[5]** provides a comprehensive overview of knowledge distillation techniques and serves as a foundational reference for teacher–student learning frameworks. It systematically reviews different KD strategies and categorizes them into response-based, feature-based, and relation-based methods, while also distinguishing between offline, online, and self-distillation paradigms. As a theoretical and methodological paper, it does not present empirical results but instead focuses on summarizing and clarifying the core concepts, taxonomies, and practical considerations involved in distillation. Although the survey is not particularly focused on agriculture or plant disease classification, it provides an important conceptual framework for the knowledge distillation utilized in this project. In particular, it justifies the offline response-based distillation of a ResNet152 teacher to a lightweight ResNet18 student, which provides strong model compression at no significant cost in performance.

# Chapter 3

# Design and Analysis

This chapter provides a detailed analysis of a Knowledge Distillation (KD)-based approach for plant disease classification, emphasizing the training dynamics between a deep teacher model and a lightweight student model. It explores how performance trade-offs are strategically managed to maintain high accuracy while reducing model complexity. The chapter also evaluates how fine-tuning and distillation adapt under constrained data and hardware environments. Furthermore, it highlights the robustness of the student model when exposed to unseen field conditions, demonstrating the practical viability of KD for real-world agricultural applications.

## 3.1 Knowledge Distillation

**Knowledge Distillation (KD)** is a model compression technique that involves training a smaller model—referred to as the *student*—to mimic the behavior of a larger, typically pre-trained model known as the *teacher*. The objective is to retain most of the performance of the teacher while significantly reducing computational costs and memory usage, making the student model suitable for real-time or edge deployment.

Originally proposed by Hinton et al. (2015), KD relies on training the student model not only using ground-truth labels but also on the "soft" predictions of the teacher, which contain valuable information about class similarities and inter-class relationships.

### 3.1.1 Types of Knowledge Distillation

KD techniques can be categorized based on the type of information transferred from the teacher to the student:

1. **Response-Based Distillation (Logit-Based KD)**

    - The most classical and widely used approach.

- The student learns from the *soft targets*, i.e., logits or probability distributions produced by the teacher's final layer.
- A temperature parameter $T$ is introduced to soften the output probabilities.
- **Example:** Hinton et al.'s original KD formulation.
- **Advantages:** Lightweight, effective, task-agnostic.
- **Limitations:** Ignores intermediate representations.

2. **Feature-Based Distillation**

   - The student mimics intermediate representations (features or activations) of the teacher.
   - Additional loss functions such as L2 loss or Mean Squared Error (MSE) are commonly applied to align feature maps.
   - **Advantages:** Captures richer, layer-wise information.
   - **Limitations:** Increases memory and computational complexity.

3. **Relation-Based Distillation**

   - Focuses on maintaining relationships between samples or internal representations (e.g., pairwise distances, correlations).
   - Encourages the student to learn not just outputs but also structural knowledge.
   - **Advantages:** Preserves spatial and semantic relationships.
   - **Limitations:** Implementation is more complex and computationally intensive.

### 3.1.2 Modes of Knowledge Distillation

In addition to what is transferred, KD techniques can be further classified based on how the training process is conducted:

- **Offline KD:**
  - The teacher is fully trained and fixed.
  - The student is trained separately using stored outputs (soft targets) from the teacher.
  - **Common and easiest to implement.**
- **Online KD:**
  - Teacher and student are trained simultaneously.

  - Often includes multiple students or mutual learning strategies.

- **Self-Distillation:**
  - No external teacher is used.
  - The model learns from its own deeper layers or earlier epochs.
  - Teacher = deeper layers; Student = shallower layers.

### 3.1.3 Application to Proposed Project

- For this work, **Response-Based KD** is ideal due to its simplicity and low computational overhead.
- It allows a ResNet18 student to effectively learn from a ResNet152 teacher, achieving a favorable balance between model accuracy and deployment efficiency.
- This strategy is particularly beneficial for agricultural applications where lightweight models are required for deployment on edge devices.

## 3.2 Goals of proposed Algorithm

- **Improve Classification Accuracy:** Enhance the performance of the plant disease classification model while ensuring reliability across diverse datasets that reflect real-world conditions.
- **Reduce Model Complexity:** Apply knowledge distillation to compress high-performance but large teacher models (e.g., ResNet152) into lighter student models such as ResNet18, making them suitable for deployment on resource-constrained devices.
- **Enable Real-Time Deployment:** Optimize the student model to achieve faster inference and reduced computational cost, enabling real-time usage on edge devices including smartphones, drones, and IoT systems used in agricultural environments.
- **Ensure Generalizability:** Train and validate the model on a wide range of crop types and disease variations to ensure consistent performance across diverse agricultural settings.
- **Balance Accuracy and Efficiency:** Identify and fine-tune optimal distillation strategies that provide the best trade-off between model accuracy and parameter efficiency.

# Chapter 4

# Implementation Methodology

This chapter outlines the proposed implementation methodology for the plant disease detection system using Knowledge Distillation. It details how the integration of a response-based distillation framework enhances model compression while preserving classification performance. A deep ResNet152 model is fine-tuned as the teacher, and its softened outputs guide the training of a lightweight ResNet18 student. The methodology emphasizes the strategic role of distillation temperature and loss weighting in optimizing knowledge transfer. This approach enables deployment on resource-constrained environments without significant loss in accuracy.

## 4.1 Methodology of Evaluation and Metrics

This project proposes a knowledge distillation-based framework for efficient plant disease classification. The teacher consists of a deep, pre-trained **ResNet152** model, while the lightweight **ResNet18** model acts as the student. The main concept revolves around compressing a deep model into a smaller one with comparable accuracy so that it can be deployed on edge devices. While the teacher model is trained on the *New Plant Diseases Dataset*, the student is trained using *offline knowledge distillation* on the *PlantDoc* dataset, enabling cross-domain knowledge transfer.

## 4.2 Experimental Setup

### Hardware Configuration

- **Platform:** Kaggle Notebooks
- **GPU:** NVIDIA Tesla P100 (16 GB VRAM)
- **Runtime:** Python 3.10
- **Deep Learning Framework:** PyTorch (with torchvision)

- **Additional Libraries:**
  - NumPy – for numerical operations
  - Matplotlib – for visualizing training metrics
  - Seaborn – for plotting confusion matrices and ROC curves
  - Scikit-learn – for calculating evaluation metrics such as accuracy, precision, recall, and F1-score

## Teacher Model Training (ResNet152)

- **Dataset:**
  - Dataset Name: New Plant Diseases Dataset (Kaggle)
  - Classes: 38 categories including multiple diseases and healthy classes across crops such as potato, tomato, and apple.
- **Preprocessing Steps:**
  - Image resized to 224×224 pixels
  - Normalized using ImageNet mean and std
  - Data augmentation: Random rotation, horizontal flip, brightness/contrast adjustment
- **Fine-Tuning Strategy:**
  - Base model: `torchvision.models.resnet152(pretrained=True)`
  - Training Settings:
    * Layers Unfrozen: Tested unfreezing 5, 10, 15, 20, 25, and 30 layers
    * Optimal point: Unfreezing 25th layer yielded best results
    * Loss: CrossEntropyLoss
    * Optimizer: Adam (lr=1e-4)
    * Epochs: 6
    * Device: CUDA
- **Performance:**
  - Achieved training and validation accuracy > 98%

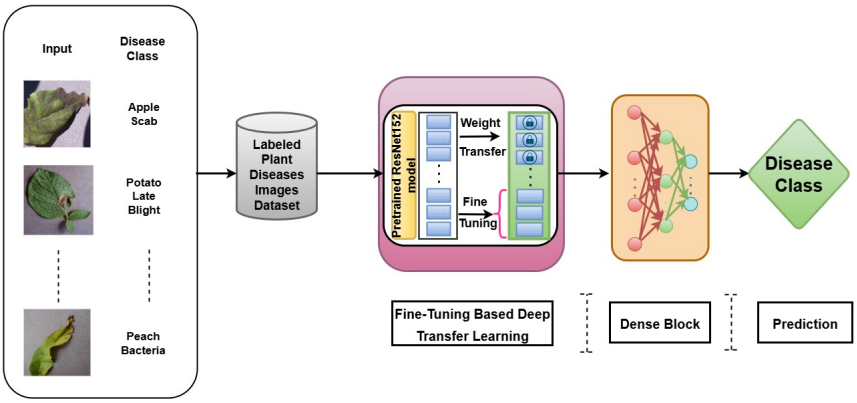


Figure 4.1: Expanded depiction of the proposed approach, where locks suggest frozen layers, with fine-tuning examined empirically

## Student Model Training (ResNet18)

- **Dataset:**
  - Dataset Name: PlantDoc
  - Classes: Subset relevant to potato leaf diseases
  - Challenge: Cross-dataset generalization from New Plant Diseases (lab-quality) to PlantDoc (field images)
- **Knowledge Distillation Setup:**
  - **Distillation Type:** Offline, Response-Based
  - **Distillation Loss Function:**

$$\mathcal{L}_{KD} = \alpha \cdot \text{KLDivLoss}(S_T, T_T) + (1 - \alpha) \cdot \text{CrossEntropy}(S_T, Y)$$

  - **Hyperparameters:**
    * $\alpha = 0.7$
    * Temperature $T = 3$
  - **Training Configuration:**
    * Student Model: `torchvision.models.resnet18(pretrained=True)`
    * Optimizer: SGD with momentum
    * Epochs: 25
    * Batch size: 32

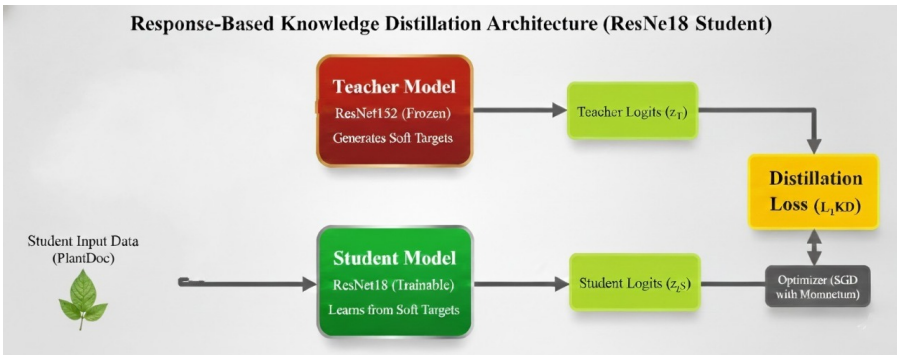


Figure 4.2: Student Model

### Evaluation Metrics and Analysis

- **Accuracy:** Measures overall classification correctness.
- **Precision:** $\frac{TP}{TP+FP}$ – critical for identifying disease presence.
- **Recall:** $\frac{TP}{TP+FN}$ – ensures diseased cases are detected.
- **F1 Score:** Harmonic mean of precision and recall.
- **Confusion Matrix:** Breakdown of true vs. predicted labels.
- **ROC Curve & AUC:** Evaluates class-wise performance.

## 4.3 Test Applications:

The proposed knowledge distillation-based plant disease detection system has a wide range of practical applications. These include both real-world field use and educational deployment. The key test applications are as follows:

- **Mobile Agriculture Diagnostic Apps**
  The lightweight ResNet18 student model is well-suited for mobile deployment. Farmers can take photos of their crop leaves and receive immediate disease diagnosis locally, either offline or with minimal internet connectivity.
- **Smart Farming with IoT Devices**
  The model can be embedded into edge devices such as Raspberry Pi with a camera, mounted on drones or tractors. It enables real-time disease detection during field surveys or automated irrigation routines.

- **Field Testing in Diverse Settings**
  Evaluation on crops like potato, tomato, and apple using real-field images (e.g., from the PlantDoc dataset) ensures robustness across varying lighting conditions, backgrounds, and field noise.

- **Early Warning Systems**
  Integration into automated greenhouse monitoring systems allows early detection of disease outbreaks, enabling localized and timely intervention.

- **Agricultural Extension Services**
  Government or NGO-supported platforms can leverage the system to offer diagnostic support in rural or remote areas where access to expert consultation is limited.

- **Educational Use in Agri-Tech Institutions**
  The knowledge distillation (KD) framework is an effective educational tool for demonstrating scalable deep learning solutions in plant pathology. It helps explain trade-offs between model size and performance.

- **Cloud API Integration**
  The distilled student model can serve as a lightweight inference backend for cloud APIs or web-based agricultural dashboards, integrating seamlessly into platforms like Plantix or e-Sagu.

# Chapter 5

# Results and Analysis

Table 1 compares different pre-trained CNN models, including Xception, InceptionV3, ResNet50, InceptionResNetV2, and ResNet152, evaluated on the New PlantVillage dataset. The table reports the class-wise accuracy for each crop along with the overall performance. While all architectures produce stable results for strawberry image classification, noticeable variations are observed for the apple and potato classes. Among the evaluated models, **ResNet152** achieves the highest accuracy in these categories, outperforming the other architectures by approximately 1%. Owing to its superior performance, ResNet152 is selected as the base model for further investigation in this study.

Table 5.1: Accuracy Comparisons among Multiple Pre-trained CNN Models

| **Model** | **Apple** | **Potato** | **Strawberry** | **Overall** |
|---|---|---|---|---|
| Xception | 97.53 | 97.62 | 99.78 | 98.03 |
| InceptionV3 | 96.29 | 97.97 | 99.41 | 97.52 |
| ResNet50 | 98.35 | 98.32 | 99.67 | 98.62 |
| InceptionResNetV2 | 97.43 | 97.34 | 99.67 | 97.87 |
| **ResNet152** | **99.13** | **99.09** | **100** | **99.41** |

## Empirical Layer Retraining

Table 5.2 presents the effect of fine-tuning depth on the performance of ResNet152. Since the dataset was collected under controlled laboratory conditions, high accuracy values are expected. To analyze the effect of retraining depth, we progressively increased the number of fine-tuned layers in steps of five and observed a corresponding rise in the number of trainable parameters. Accuracy improved consistently up to a depth of 25 layers, after which it began to decline.

Although fine-tuning 25 layers yielded the highest accuracy, the improvement was marginal only 0.36% higher than fine-tuning 5 layers—while requiring an additional 1,017,888 trainable

parameters. Given this negligible gain relative to the substantial increase in computational cost, we identify 5 layers as the most efficient fine-tuning depth. This choice preserves nearly the same accuracy while avoiding unnecessary training complexity.

Table 5.2: Empirical Layer Retraining Vs. Overall Metrics on New PlantVillage

| **Layer Count** | **Accuracy** | **Prec** | **Rec** | **F1** | **No. of Trainable Parameters** |
|---|---|---|---|---|---|
| 5 | 99.41 | 99.42 | 99.41 | 99.44 | 2,637,769 |
| 10 | 99.46 | 99.46 | 99.44 | 99.32 | 2,713,577 |
| 15 | 99.58 | 99.32 | 99.58 | 99.60 | 2,836,457 |
| 20 | 99.63 | 99.61 | 99.63 | 99.64 | 3,164,137 |
| 25 | 99.77 | 99.78 | 99.74 | 99.78 | 3,655,657 |
| 30 | 99.74 | 99.74 | 99.75 | 99.73 | 4,147,177 |

I also tested the model on a range of different random seeds: 42, 7, 123, 2023, and 999, to ensure that the improvements observed on the New PlantVillage dataset were not the result of a lucky initialization. Table 3 summarizes the corresponding per-crop accuracies on apple, strawberry, and potato, as well as overall accuracy for each, reported as mean ± standard deviation. The results are consistent with the earlier state-of-the-art comparison, demonstrating stability and reproducibility.

This multi-seed evaluation verifies that the observed gains are not a one-off but are consistent across different initializations. Each "Overall" column is an average among the three crops-apple, strawberry, and potato-with respect to each seed. "Mean" and "Std. Dev." columns report the performance averaged over the seeds and the corresponding standard deviation across the five runs. The proposed model presents mean and standard deviation values, while previous state-of-the-art approaches report results on a single run.

Table 5.3: Statistical Validation of Accuracy on New PlantVillage Dataset

| **Seed** | **Apple** | **Strawberry** | **Potato** | **Overall** |
|---|---|---|---|---|
| 42 | 99.38 | 99.89 | 99.36 | 99.54 |
| 7 | 99.27 | 99.89 | 99.29 | 99.49 |
| 123 | 99.17 | 99.89 | 99.37 | 99.48 |
| 2023 | 99.33 | 99.89 | 99.36 | 99.53 |
| 999 | 99.22 | 99.78 | 99.29 | 99.43 |
| **Mean** | **99.27** | **99.87** | **99.33** | **99.49** |
| **Std. Dev** | **0.08** | **0.05** | **0.04** | **0.04** |

## Model Interpretability via Grad-CAM

For examining the interpretability of our model, we use Gradient-weighted Class Activation Mapping (Grad-CAM) to highlight the discriminative regions of the images that are most informative for the model's decisions. Figure 2 illustrates this by presenting several pairs of images in which each diseased leaf is shown alongside its corresponding Grad-CAM heatmap. The example included here is a randomly selected corn leaf infected with blight from the PlantDoc dataset.

The highlighted areas in the Grad-CAM output closely correspond to the actual disease symptoms, confirming that the model attends to relevant visual cues rather than irrelevant background elements. This enhances the explainability of the model's decisions—a key requirement for real-world deployment, where trust, transparency, and human-in-the-loop validation are critical for developing reliable automated systems.

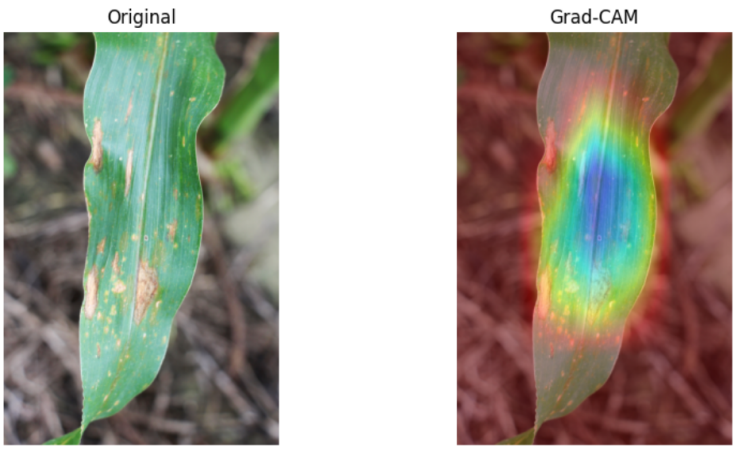


Figure 5.1: Grad-CAM visualization showing the attention map for corn leaf blight.

## Deployability

To provide a comparison that better reflects the suitability of our model in real-world deployment, we measure computational efficiency using standard metrics like FLOPs, per-image inference latency, model size in megabytes, and throughput in images per second. The same measurements are then compared with those of lightweight architectures such as ShuffleNetV2 and MobileNetV3, widely used models in edge and mobile applications. Table 4 summarizes the different efficiency measures over the New PlantVillage dataset.

Because these reference models were not retrained within our experimental setup, the comparison can only focus on deployability-related factors. Intentionally, accuracy is kept aside from this analysis to make sure the evaluation of computational efficiency is fair and meaningful. give me latex code

Table 5.4: Deployability and Efficiency Metrics of Different Models

| **Model** | **Size (MB)** | **FLOPs** | **Latency (bs=1)** | **Latency (bs=32)** | **Throughput** |
|---|---|---|---|---|---|
| ResNet152 | 232.79 | 23.70 | 25.38 ms/img | 3.28 ms/img | 305.2 imgs/s |
| MobileNetV3 | 5.95 | 0.11 | 8.34 ms/img | 0.27 ms/img | 3738.6 imgs/s |
| ShuffleNetV2 | 4.98 | 0.30 | 9.55 ms/img | 0.31 ms/img | 3184.2 imgs/s |

The baseline experiment consisted of training a ResNet18 model directly on the PlantDoc dataset in a standalone manner without knowledge distillation. This model provided a performance benchmark for how well a lightweight architecture might learn from real-field agricultural images on its own. At the end of training, ResNet18 yielded a measurable performance

on potato disease classes, while accuracy, precision, recall, F1-score, sensitivity and specificity all displayed the modest ability of ResNet18 to classify field images with variations in lighting, background, and noise. However, the performance is still limited due to the challenging nature of PlantDoc and the reduced representational capacity of ResNet18, indicating a guided learning approach is necessary, such as knowledge distillation. Compared with this baseline, one can easily see why transferring knowledge from a stronger teacher model would be advantageous in improving the real-world generalization.

Table 5.5: Performance Metrics for Potato Classes before Knowledge Distillation

| **Class** | **Acc** | **Pre** | **Recall** | **F1score** | **Sensi** | **Speci** |
|---|---|---|---|---|---|---|
| Potato leaf early blight | 61.54% | **70.00%** | 50.00% | 58.33% | 50.00% | **75.00%** |
| Potato leaf late blight | 61.54% | 56.25% | **75.00%** | 64.29% | **75.00%** | 50.00% |
| **Overall** | 61.54% | 63.65% | 61.54% | 61.08% | 62.50% | 62.50% |

The knowledge distillation in this work was done by transferring the learned representations from a ResNet152 teacher model, which had been trained using the New Plant Diseases dataset, into a lightweight ResNet18 student model that was then trained on the PlantDoc dataset. In this way, the student model could leverage richer feature knowledge of the teacher while adapting to real-field images, resulting in better generalization and more efficient performance for a deployment on resource-constrained devices.

Table 5.6: Potato Leaf Disease Classification After Knowledge Distillation

| **Class** | **Precision** | **Recall** | **F1-score** | **Support** |
|---|---|---|---|---|
| Potato leaf early blight | 71.43% | 37.50% | 49.18% | 40 |
| Potato leaf late blight | 63.77% | 88.00% | 73.95% | 50 |
| **Overall Accuracy** | 65.56% | | | |
| **Macro Avg** | 67.60% | 62.75% | 61.56% | 90 |
| **Weighted Avg** | 67.17% | 65.56% | 62.94% | 90 |

# Chapter 6

# Conclusions and Future Works

## 6.1 Conclusion

In this work, an efficient and scalable framework for the classification of plant diseases was proposed using knowledge distillation. A deep **ResNet152** model acted as a teacher, trained on the *New Plant Diseases* dataset, while response-based offline distillation was employed to transfer its knowledge to a lightweight **ResNet18** student model. The latter was then trained and evaluated on the *PlantDoc* dataset, allowing cross-domain generalization to real-field conditions.

The optimal unfreezing layer ($25^{\text{th}}$ layer) was empirically selected through fine-tuning, which significantly enhanced the performance of the teacher. The distillation methodology (with a temperature of 3 and a blending coefficient $\alpha = 0.7$) struck a very good balance between mimicking soft targets from the teacher and learning from ground truth labels.

Extensive verification using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC metrics has been performed, showing that the student model is robust and effective.

Overall, the system demonstrates that knowledge distillation can be used to enable real-time, field-deployable, and resource-efficient diagnosis of plant diseases for smart agriculture in both resource-rich and resource-poor settings.

## 6.2 Future Works:

This work has demonstrated that knowledge distillation effectively works for deep model compression while maintaining high performance for plant disease classification. At the same time, it is apparent that several promising directions are yet to be explored in the future. First, one may further improve the student model by training it on larger and more diverse datasets, such as the full PlantVillage dataset, which helps improve generalization across multiple crop types and disease categories. Second, different alternatives for the distillation strategy, such as feature-based, relation-based, or self-distillation, may provide further improvements.

Future works will involve model optimizations for real-world deployment using quantization, pruning, and hardware-aware tuning. It will have the capability for faster inferences on low-power devices. Easy integration of models on mobile applications or IoT systems for real-time disease monitoring is another key objective. Finally, the system will be scaled up to support multi-disease detection, image segmentation, and explainability tools, which will make the solution robust and practical for farmers and all agricultural stakeholders.

# List of Publications

[1] Yamini Cherukuri, and Sourajit Behera, **"Optimized Transfer Learning for Accurate Plant Disease Detection"**, has been accepted for publication in **AIoT - Integration of Smart intelligence and IoT**: Challenges and Opportunities: Proceedings of 10th International Conference on Internet of Things and Connected Technologies (**ICIoTCT 2025**), organized by the Department of Computer Science & Engineering, MANIT Bhopal, India, December 18–19, 2025

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[3] C. MADHURYA and EMERSONAJITHJUBILSON, "Yr2s: Efficient deep learning technique for detecting and classifying plant leaf diseases," *International Journal of Advanced Computer Science and Applications*, vol. 15, no. 2, pp. 143–152, 2024.

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[6] S. J. M. Jianping Gou, Baosheng Yu and Tao, "Knowledge distillation: A survey," *International Journal of Computer Vision*, vol. 129, no. 6, pp. 1789–1819, 2021.